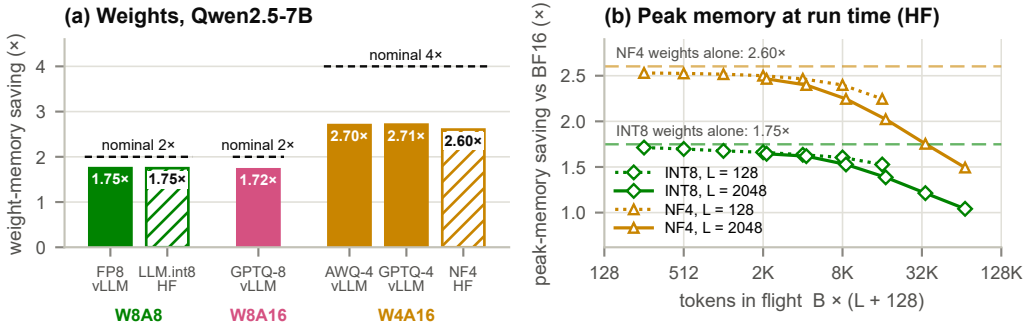
(a) Weights, Qwen2.5-7B
weight-memory saving (×)
0
1
2
3
4
nominal 2×
nominal 2×
nominal 4×
1.75×
1.75×
1.72×
2.70×
2.71×
2.60×
FP8 vLLM
LLM.int8 HF
GPTQ-8 vLLM
AWQ-4 vLLM
GPTQ-4 vLLM
NF4 HF
W8A8
W8A16
W4A16
(b) Peak memory at run time (HF)
peak-memory saving vs BF16 (×)
1.0
1.5
2.0
2.5
NF4 weights alone: 2.60×
INT8 weights alone: 1.75×
INT8, L = 128
INT8, L = 2048
NF4, L = 128
NF4, L = 2048
128
512
2K
8K
32K
128K
tokens in flight B × (L + 128)